



FUZZY LOGIC



Source:
Intelligent Control Systems with LabVIEW™
Chapter 2
Fuzzy Logic

2.1 Introduction

- The real world is **complex**; this complexity generally arises from **uncertainty**.
- Humans have unconsciously been able to address complex, ambiguous, and uncertain problems thanks to the gift of thinking.
- This thought process is possible because humans do not need the complete description of the problem since they have the capacity to **reason approximately**.
- With the advent of computers and their increase in computation power, engineers and scientists are more and more interested in the creation of methods and techniques that will allow **computers to reason with uncertainty**.

Machine can't!!! WHY???

Classical set theory is based on the fundamental concept of a *set*, in which individuals are either a member or not a member. A sharp, crisp, and ambiguous distinction exists between a member and a non-member for any well-defined set of entities in this theory, and there is a very precise and clear boundary to indicate if an entity belongs to a set. Thus, in classical set theory an element is not allowed to be in a set (1) or not in a set (0) at the same time. This means that many real-world problems cannot be handled by classical set theory. On the contrary, the fuzzy set theory accepts partial membership values $\mu_f \in [0, +1]$, and therefore, in a sense generalizes the classical set theory to some extent.

Machine can be a solution???

As Prof. Lotfi A. Zadeh suggests by his *principle of incompatibility*: “The closer one looks at a real-world problem, the fuzzier becomes the solution,” and thus, imprecision and complexity are correlated [1]. Complexity is inversely related to the understanding we can have of a problem or system. When little complexity is presented, closed-loop forms are enough to describe the systems. More complex systems need methods such as neural networks that can reduce some uncertainty. When systems are complex enough that only few numerical data exist and the majority of this information is vague, fuzzy reasoning can be used for manipulating this information.

2.2 Industrial Applications

Example 2.1. Mitsubishi manufactures a **fuzzy air conditioner**. While conventional air conditioners use on/off controllers that work and stop working based on a range of temperatures, the Mitsubishi machine takes advantage of fuzzy rules; the machine operates smoother as a result. The machine becomes mistreated by sudden changes of state, more consistent room temperatures are achieved, and less energy is consumed. These were first released in 1989. □

Example 2.2. Fisher, Sanyo, Panasonic, and Canon make **fuzzy video cameras**. These have a digital image stabilizer to remove hand jitter, and the video camera can determine the best focus and lightning. Fuzzy decision making is used to control these actions. The present image is compared with the previous frame in memory, stationary objects are detected, and its shift coordinates are computed. This shift is subtracted from the image to compensate for the hand jitter. □

Example 2.3. Fujitec and Toshiba have a fuzzy scheme that evaluates the passenger traffic and the elevator variables to determine car announcement and stopping time. This helps reduce the waiting time and improves the efficiency and reliability of the systems. The patent for this type of system was issued in 1998. □

Example 2.4. The automotive industry has also taken advantage of the theory. Nissan has had an **anti-lock braking** system since 1997 that senses wheel speed, road conditions, and driving pattern, and the fuzzy ABS determines the braking action, with skid control [3]. □

Example 2.5. Since 1988 Hitachi has turned over the control of the Sendai subway system to a fuzzy system. It has reduced the judgment on errors in acceleration and braking by 70%. The Ministry of International Trade and Industry estimates that in 1992 Japan produced about \$2 billion worth of fuzzy products. US and European companies still lag far behind. The market of products is enormous, ranging from fuzzy toasters to fuzzy golf diagnostic systems. □

2.3.1 Uncertainty in Information

- The uncertainties in a problem should be carefully studied by engineers prior to selecting an appropriate method to represent the uncertainty and to solve the problem.
- Fuzzy sets provide a way that is very similar to the human reasoning system.
- In universities most of the material taught in engineering classes is based on the presumption that **knowledge is deterministic**.
- Then when students graduate and enter “the real world,” they fear that they will forget the correct formula.
- However, one must realize that **all information contains a certain degree of uncertainty**.
- Uncertainty can arise from many factors, such as complexity, randomness, ignorance, or imprecision.
- We all use uncertain information and imprecision to solve problems.
- Hence, our computational methods should be able to represent and manipulate fuzzy and statistical uncertainties.

2.3.2 Concept of Fuzziness

- In our everyday language we use a great deal of **vagueness and imprecision**, that can also be called fuzziness.
- **Some examples** are: a person's size is *tall*, and their age is classified as *young*.
- Terms such as *tall* and *young* are fuzzy because they cannot be crisply defined, although as humans we use this information to make decisions.
- When we want to classify a person as tall or young it is impossible to decide if the person is in a set or not.

Buying a used car

Consider the set of cars for sale in your area.

Let A be the fuzzy set of cars that you could consider owning.

For each car you can assign it a value between 0 and 1 which would represent the degree of membership of this particular car in the fuzzy set A .

For a given car this depends on its age, condition, style, price and so forth. Here the fuzzy membership function measures the overall attractiveness of a car to you.

Why Fuzzy Sets

- It enables one to work in uncertain and ambiguous situations and solve ill-posed problems or problems with incomplete information

- Example

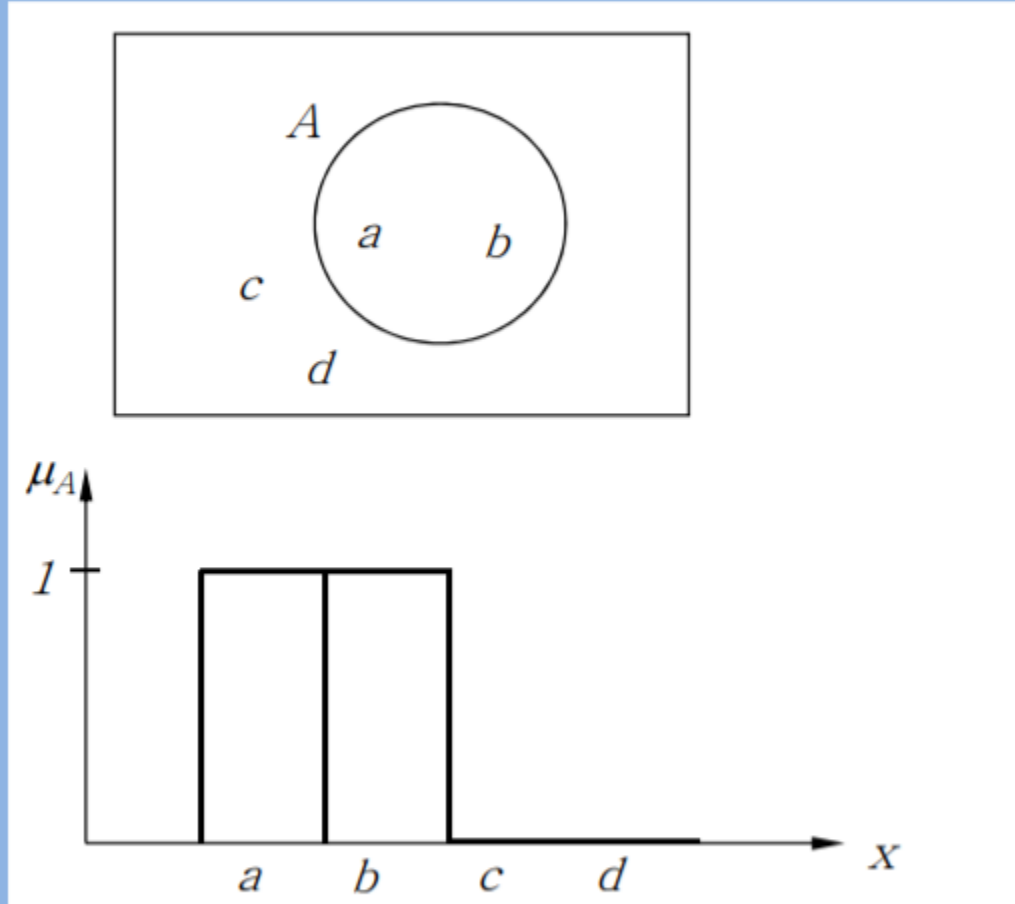


Fig : Graphical representation of crisp set

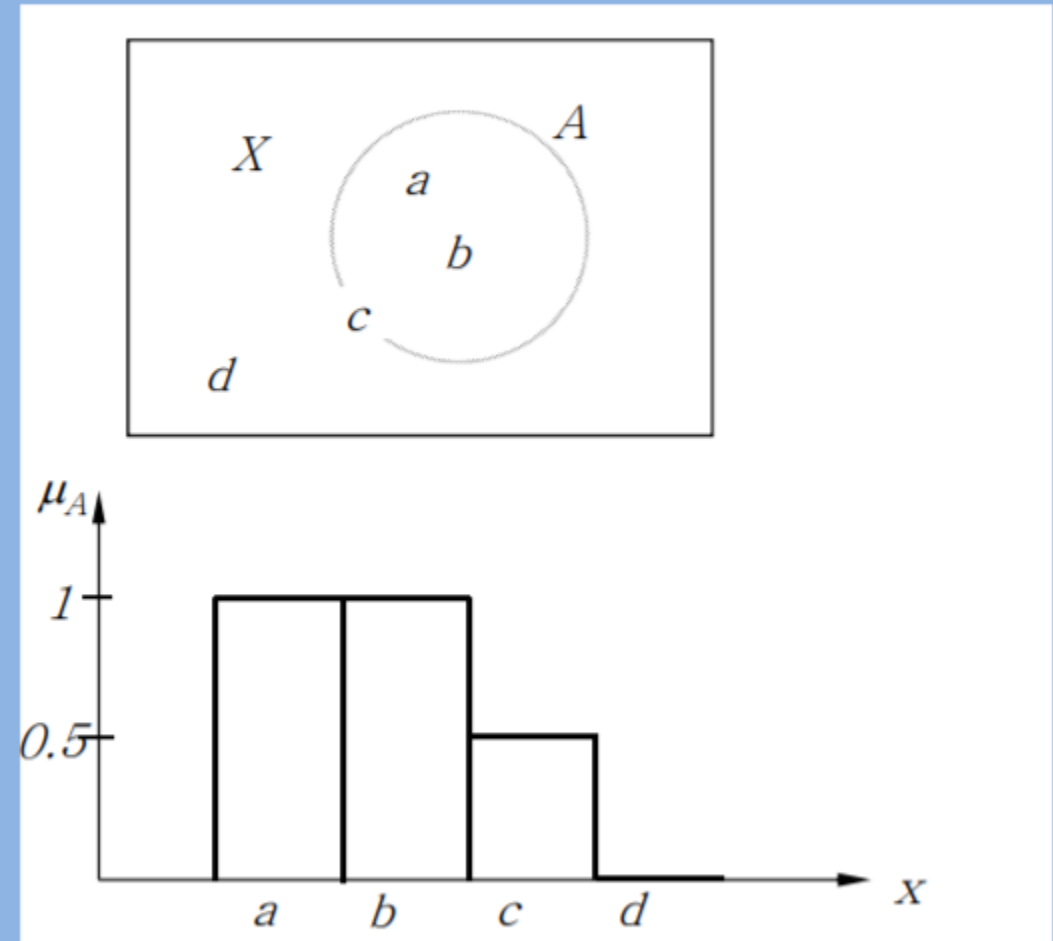


Fig : Graphical representation of fuzzy set

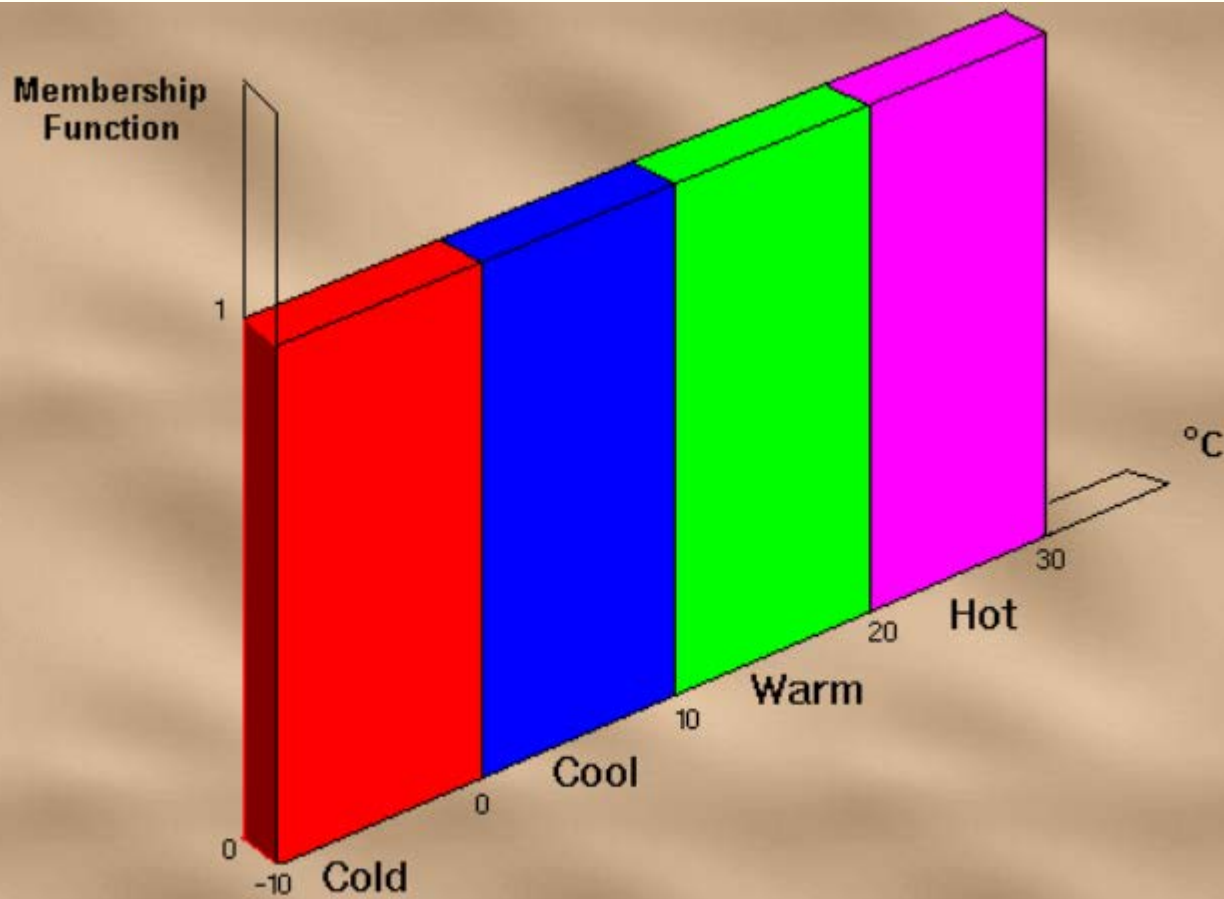


Fig. 1 : Bivalent Sets to Characterize the Temp. of a room.

it is not accurate to define a transition from a quantity such as 'warm' to 'hot' by the application of one degree Fahrenheit of heat. In the real world a smooth (unnoticeable) drift from warm to hot would occur.

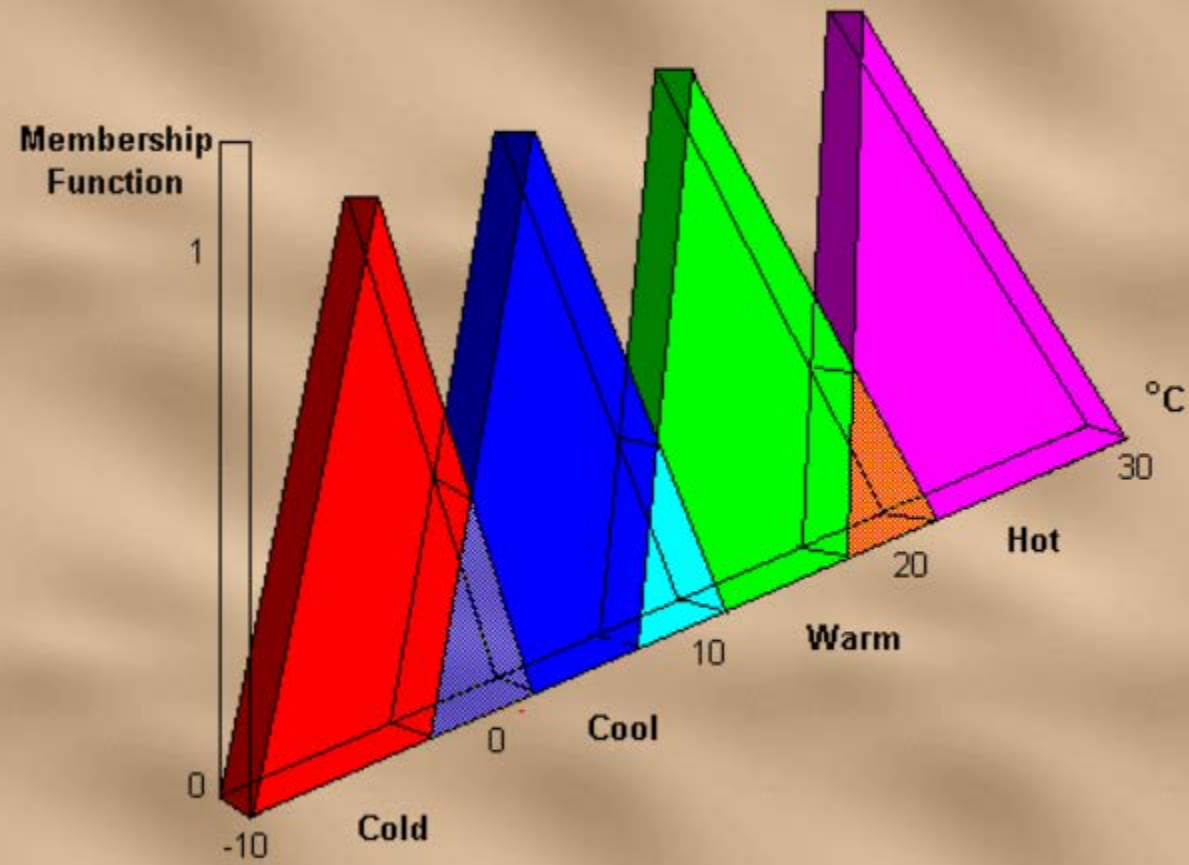


Fig. 2 - Fuzzy Sets to characterize the Temp. of a room.

Let's talk about **people** and "**youthness**".

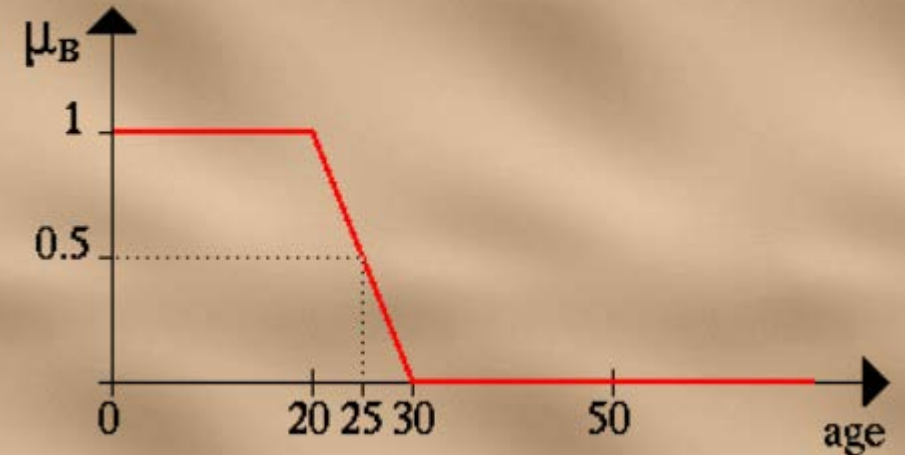
In this case the set S (the universe of discourse) is the set of people. A fuzzy subset YOUNG is also defined, which answers the question "to what degree is person x young?"

Given this definition, here are some example values:

Person	Age	degree of youth
Johan	10	1.00
Edwin	21	0.90
Parthiban	25	0.50
Arosha	26	0.40
Chin Wei	28	0.20
Rajkumar	83	0.00

$$\text{young}(x) = \left\{ \begin{array}{ll} 1, & \text{if } \text{age}(x) \leq 20, \\ (30 - \text{age}(x))/10, & \text{if } 20 < \text{age}(x) \leq 30, \\ 0, & \text{if } \text{age}(x) > 30 \end{array} \right\}$$

A graph of this looks like:



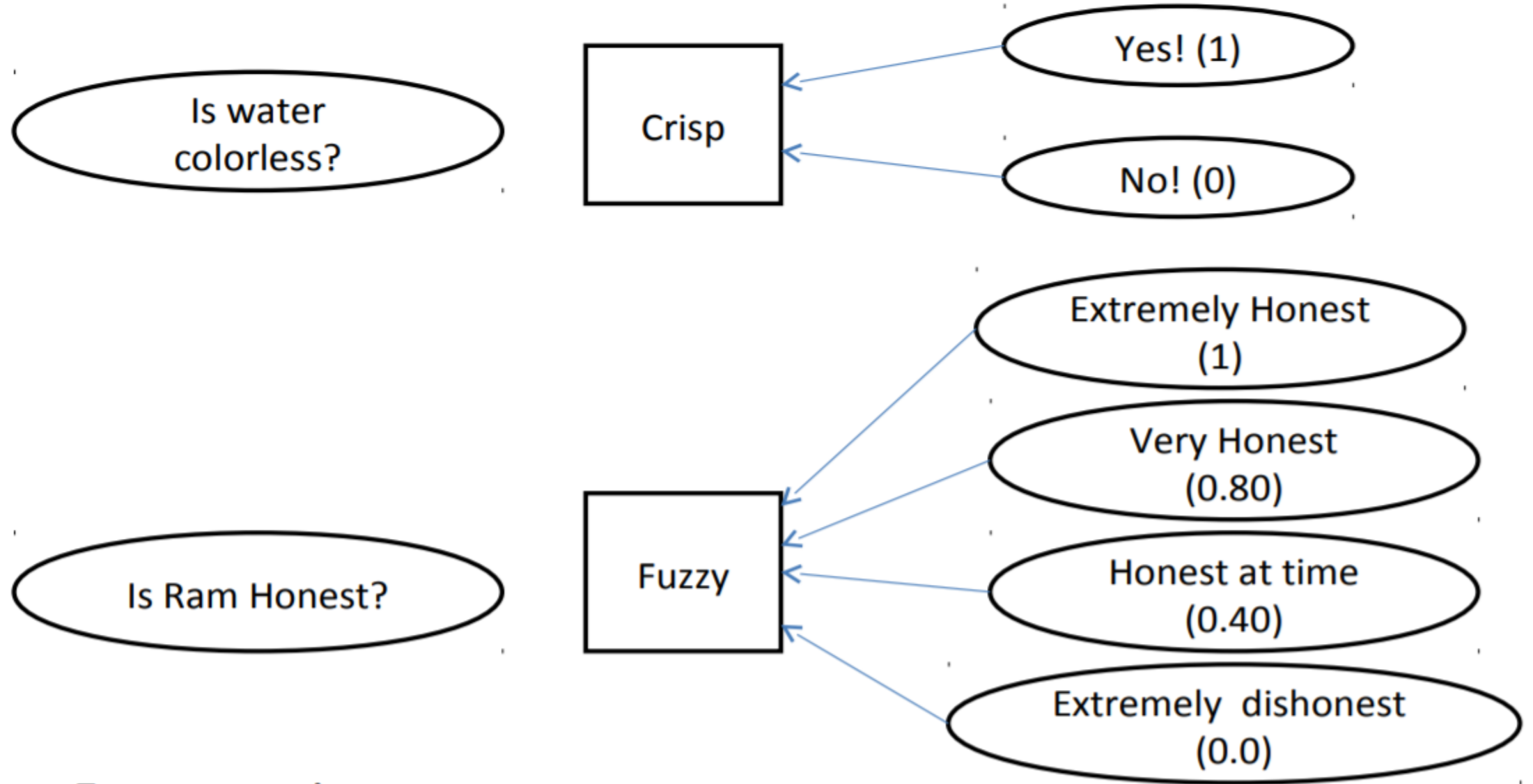
Classical set theory

- Classes of objects with sharp boundaries.
- A classical set is defined by crisp(exact) boundaries, i.e., there is no uncertainty about the location of the set boundaries.
- Widely used in digital system design

Fuzzy set theory

- Classes of objects with un-sharp boundaries.
- A fuzzy set is defined by its ambiguous boundaries, i.e., there exists uncertainty about the location of the set boundaries.
- Used in fuzzy controllers.

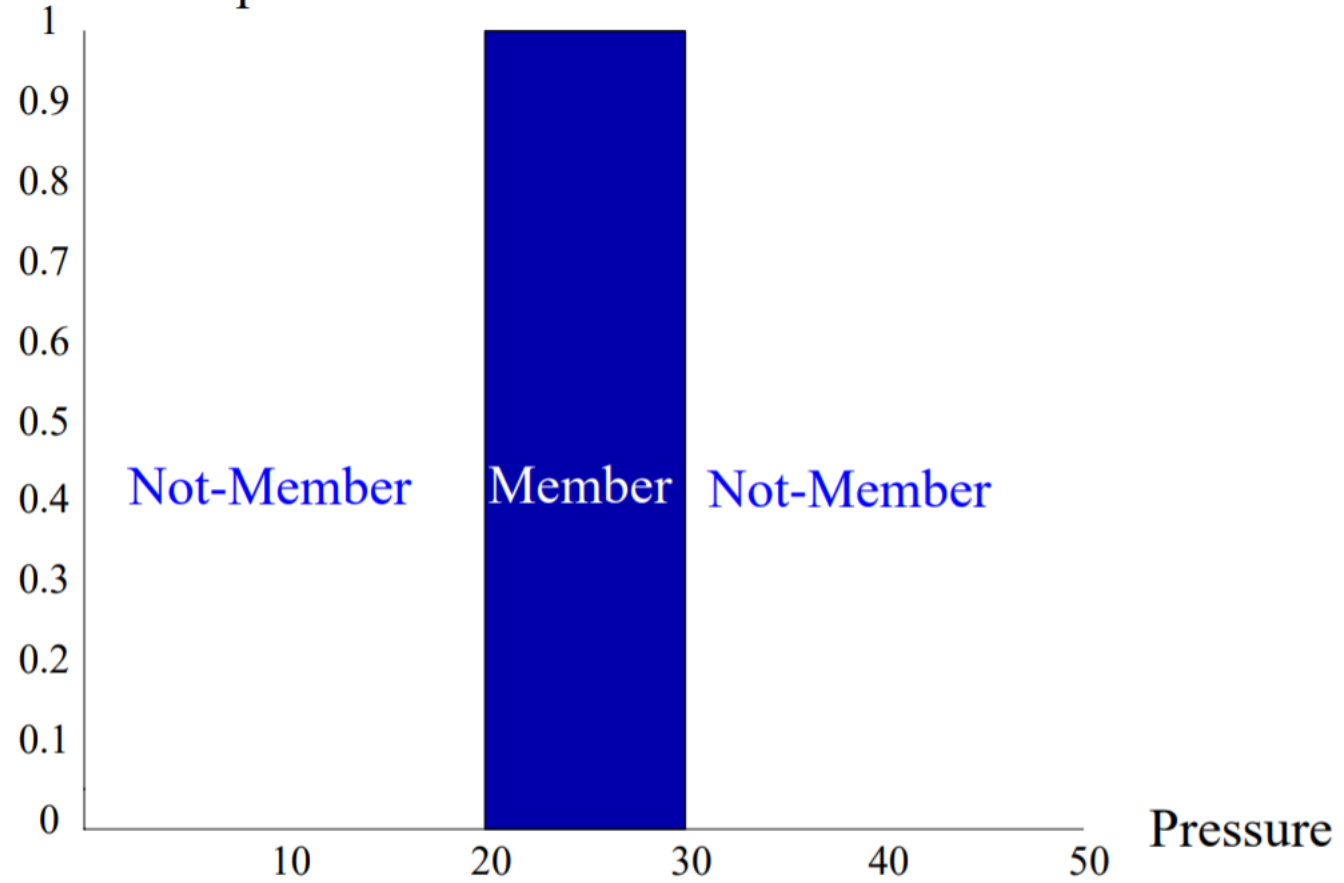
Example



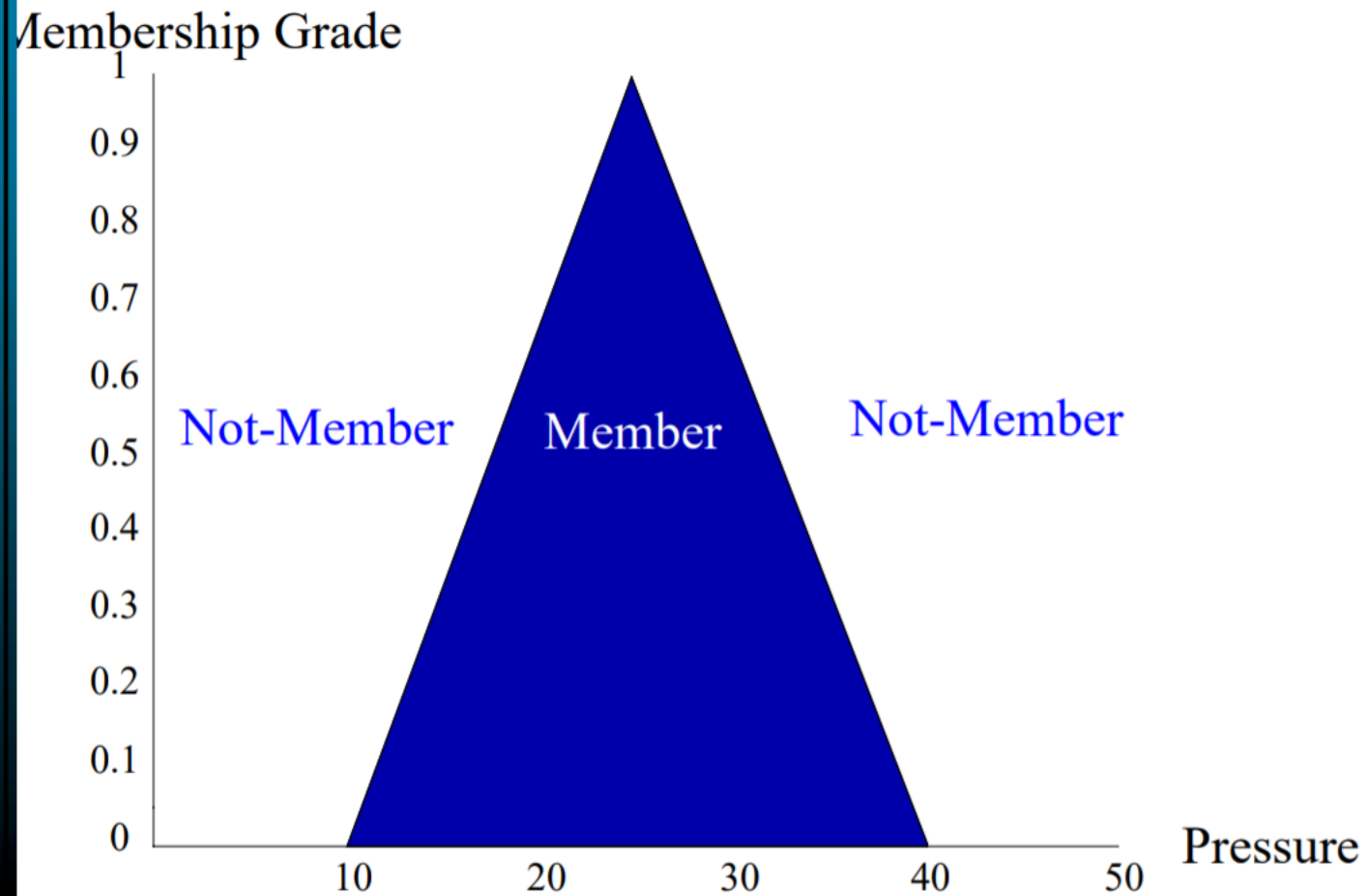
Fuzzy vs crips

Boolean Medium Pressure

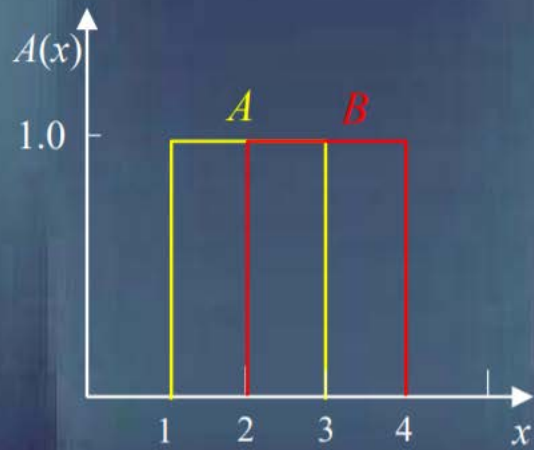
Membership Grade



Fuzzy Medium Pressure

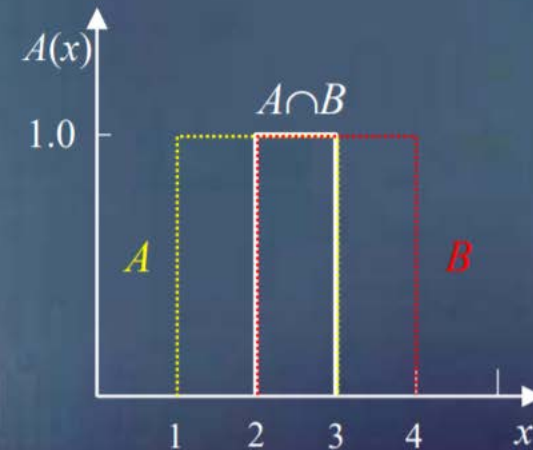


Intersection of sets



$$A = \{x \in \mathbf{R} \mid 1 \leq x \leq 3\}$$

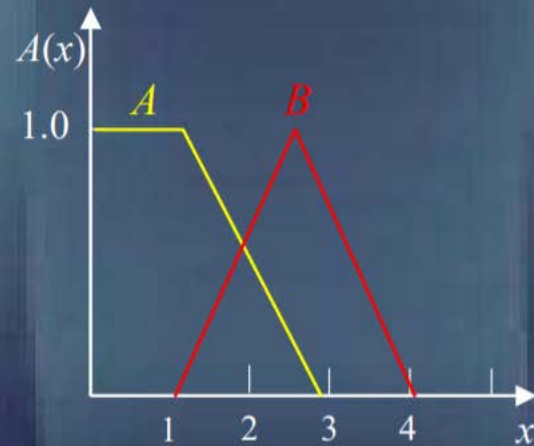
$$B = \{x \in \mathbf{R} \mid 2 \leq x \leq 4\}$$



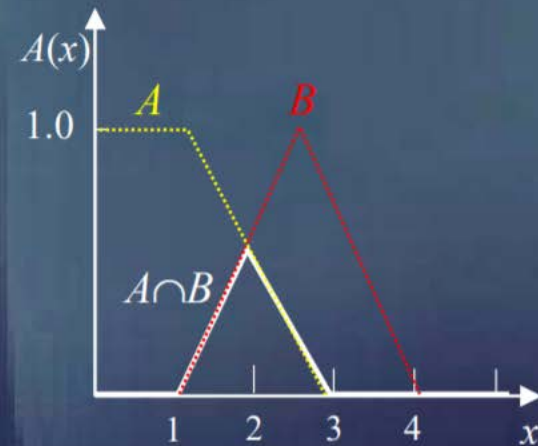
$$A \cap B: \{x \in \mathbf{R} \mid 2 \leq x \leq 3\}$$

$$(A \cap B)(x) = \min [A(x), B(x)] \quad \forall x \in \mathbf{X}$$

Intersection of fuzzy sets

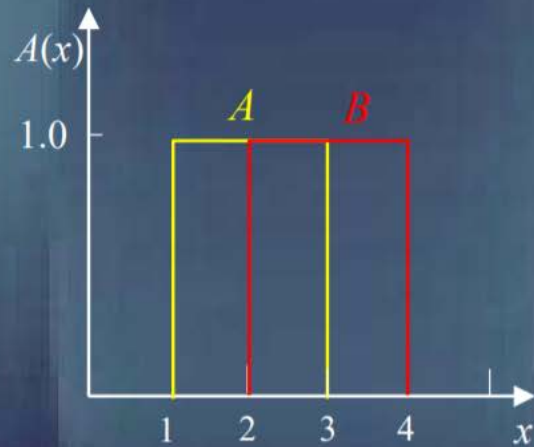


$$(A \cap B)(x) = \min [A(x), B(x)] \quad \forall x \in \mathbf{X}$$



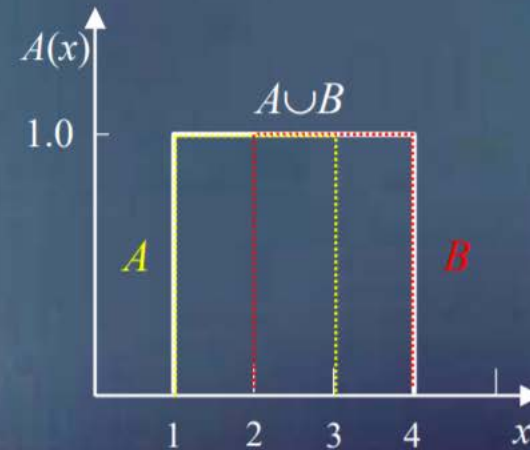
Standard intersection

Union of sets



$$A = \{x \in \mathbf{R} \mid 1 \leq x \leq 3\}$$

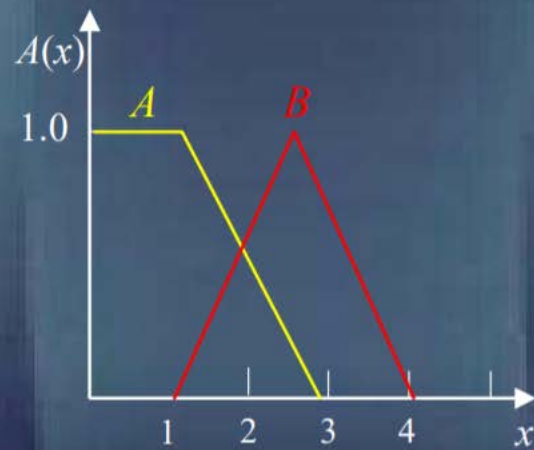
$$B = \{x \in \mathbf{R} \mid 2 \leq x \leq 4\}$$



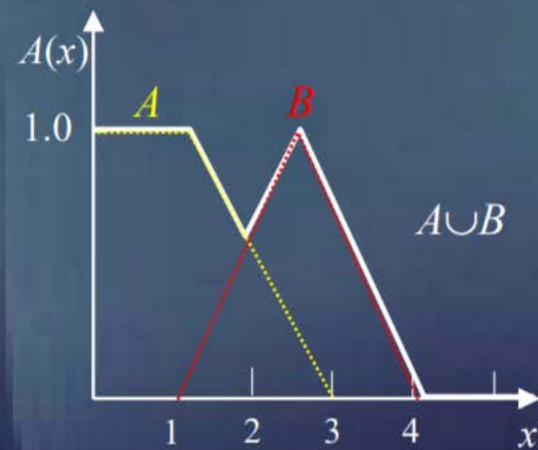
$$A \cup B: \{x \in \mathbf{R} \mid 1 \leq x \leq 4\}$$

$$(A \cup B)(x) = \max [A(x), B(x)] \quad \forall x \in \mathbf{X}$$

Union of fuzzy sets



$$(A \cup B)(x) = \max [A(x), B(x)] \quad \forall x \in \mathbf{X}$$

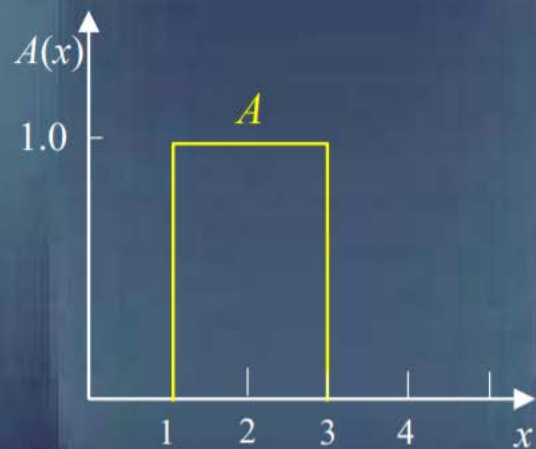


Standard union

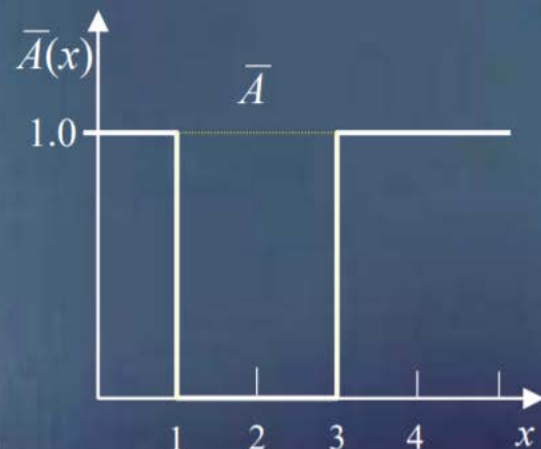
Operations with fuzzy sets: examples

1. Determine the union and intersection of the fuzzy sets $\tilde{A} =$ “comfortable house for a 4 persons - family” and $\tilde{B} =$ “small house”, where
 $\tilde{A} = \{(1, 0.1), (2, 0.5), (3, 0.8), (4, 1.0), (5, 0.7), (6, 0.2)\}$ and
 $\tilde{B} = \{(1, 1), (2, 0.8), (3, 0.4), (4, 0.1)\}$:
 $\tilde{A} \cup \tilde{B} = \{(1, \max(0.1, 1)), (2, \max(0.5, 0.8)), (3, \max(0.8, 0.4)), (4, \max(1, 0.1)), (5, \max(0.7, 0)), (6, \max(0.2, 0))\} =$
 $\{(1, 1), (2, 0.8), (3, 0.8), (4, 1), (5, 0.7), (6, 0.2)\}$
 $\tilde{A} \cap \tilde{B} = \{(1, \min(0.1, 1)), (2, \min(0.5, 0.8)), (3, \min(0.8, 0.4)), (4, \min(1, 0.1)), (5, \min(0.7, 0)), (6, \min(0.2, 0))\} =$
 $\{(1, 0.1), (2, 0.5), (3, 0.4), (4, 0.1), (5, 0), (6, 0)\}$
 $\tilde{A} \cup \tilde{B}$ can be read as “comfortable house for a 4 persons - family or small”, and $\tilde{A} \cap \tilde{B}$ as “comfortable house for a 4 persons - family and small”

Complement of sets



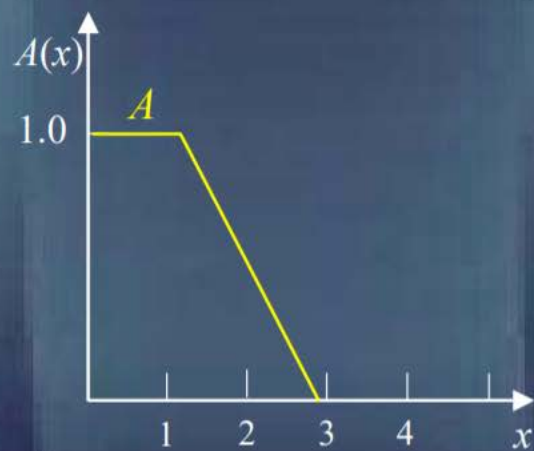
$$A = \{x \in \mathbf{R} \mid 1 \leq x \leq 3\}$$



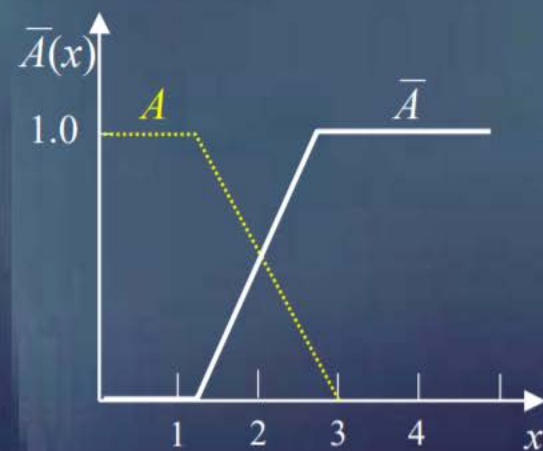
$$\bar{A}(x) = \{x \in \mathbf{R} \mid x < 1, x > 3\}$$

$$\bar{A}(x) = 1 - A(x) \quad \forall x \in \mathbf{X}$$

Complement of fuzzy sets



$$\bar{A}(x) = 1 - A(x) \quad \forall x \in \mathbf{X}$$

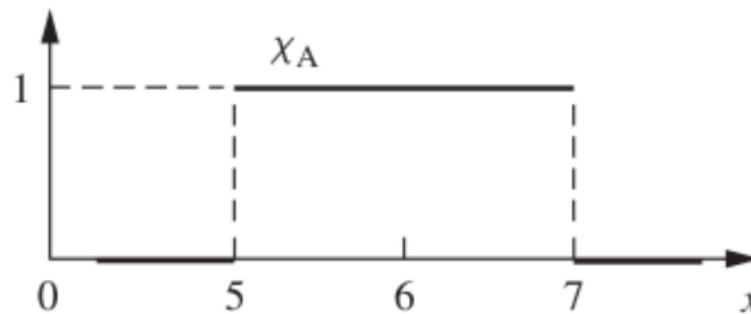
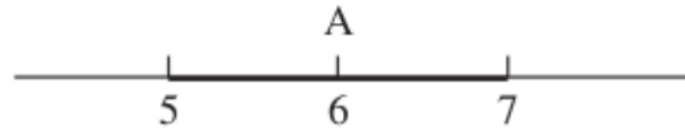


Standard complement

Introduction: Crisp Set

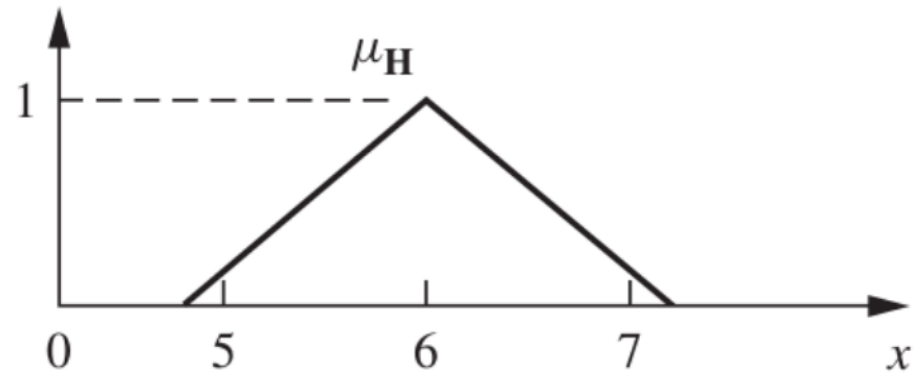
- Classical sets contain objects that satisfy precise properties of membership. For example, the set of heights *from 5 to 7 feet* is precise (crisp);

$$\chi_A(x) = \begin{cases} 1, & x \in A \\ 0, & x \notin A \end{cases}$$



Fuzzy Sets

- Consider a set H consisting of heights *near 6 feet*
- Since the property *near 6 feet* is **fuzzy**, there is no unique membership function for H , rather we have to accommodate various “degrees of membership” on the real continuous interval $[0, 1]$



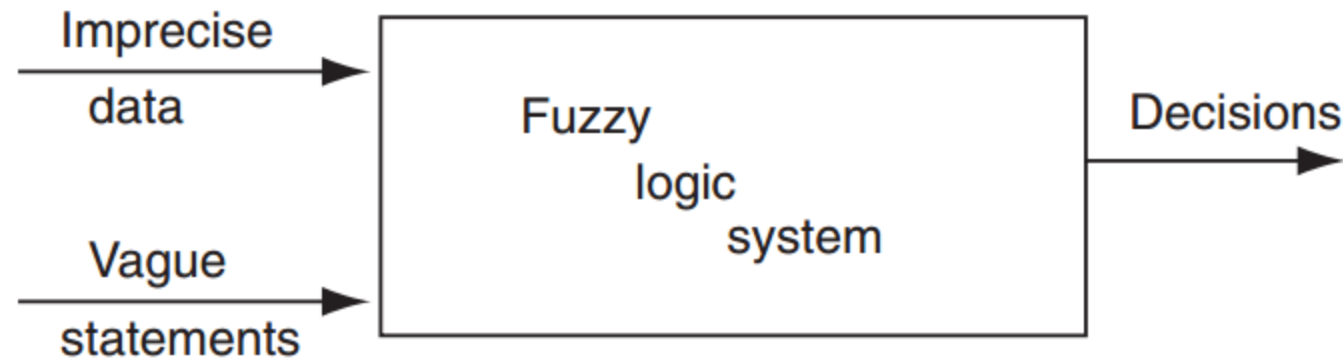


Fig. 1.1. A fuzzy logic system which accepts imprecise data and vague statements such as low, medium, high and provides decisions

Example 2.5. We want to compare two sensors based upon their detection levels and gain settings. The following table of gain settings and sensor detection levels with a standard item being monitored provides typical membership values to represents the detection levels for each of the sensors.

Gain setting	Sensor 1 detection levels	Sensor 2 detection levels
0	0	0
20	0.5	0.35
40	0.65	0.5
60	0.85	0.75
80	1	0.90
100	1	1

The universe of discourse is $X = \{0, 20, 40, 60, 80, 100\}$. Find the membership function for the two sensors: Find the following membership functions using standard set operations:

- | | |
|---|---|
| (a) $\mu_{S_1 \cup S_2}(x)$ | (b) $\mu_{S_1 \cap S_2}(x)$ |
| (c) $\mu_{\tilde{S}_1}(x)$ | (d) $\mu_{\tilde{S}_2}(x)$ |
| (e) $\mu_{\overline{S_1 \cup S_1}}(x)$ | (f) $\mu_{\overline{S_1 \cup S_1}}(x)$ |
| (g) $\mu_{\tilde{S_1} \cup \tilde{S_1}}(x)$ | (h) $\mu_{\tilde{S_1} \cup \tilde{S_1}}(x)$ |

Solution. The form are

(a) $\mu_{\tilde{S}_1 \cup \tilde{S}_2}(x) = \mu_{\tilde{s}_s}(x) \vee \mu_{\tilde{s}_2}(x)$ and discrete

$$= \left\{ \frac{0.5}{20} + \frac{0.65}{40} + \frac{0.85}{60} + \frac{1}{80} + \frac{1}{100} \right\}$$

(b) $\mu_{\tilde{S}_1 \cap \tilde{S}_2}(x) = \mu_{\tilde{s}_s}(x) \wedge \mu_{\tilde{s}_2}(x)$

$$= \left\{ \frac{0.35}{20} + \frac{0.5}{40} + \frac{0.75}{60} + \frac{0.9}{80} + \frac{1}{100} \right\}$$

(c) $\mu_{\bar{S}_1}(x) = \left\{ \frac{0.5}{20} + \frac{0.35}{40} + \frac{0.15}{60} + \frac{0}{80} + \frac{0}{100} \right\}$

(d) $\mu_{\bar{S}_2}(x) = \left\{ \frac{0.65}{20} + \frac{0.5}{40} + \frac{0.25}{60} + \frac{0.1}{80} + \frac{0}{100} \right\}$

(e) $\mu_{\overline{\tilde{S}_1 \cup \tilde{S}_1}}(x) = \mu_{\bar{S}_1 \cap \bar{S}_1} = \mu_{\bar{S}_1}(x) \wedge \mu_{\bar{S}_1}(x) = \left\{ \frac{0.5}{20} + \frac{0.35}{40} + \frac{0.15}{60} + \frac{0}{80} + \frac{0}{100} \right\}$

(f) $\mu_{\overline{\tilde{S}_1 \cap \tilde{S}_1}}(x) = \mu_{\bar{S}_1 \cup \bar{S}_1} = \mu_{\bar{S}_1}(x) \vee \mu_{\bar{S}_1}(x) = \left\{ \frac{0.5}{20} + \frac{0.35}{40} + \frac{0.15}{60} + \frac{0}{80} + \frac{0}{100} \right\}$

(g) $\mu_{\tilde{S}_1 \cup \bar{S}_1}(x) = \mu_{S_1}(x) \vee \mu_{\bar{S}_1}(x) = \left\{ \frac{0.5}{20} + \frac{0.65}{40} + \frac{0.85}{60} + \frac{1}{80} + \frac{1}{100} \right\}$

(h) $\mu_{\tilde{S}_1 \cap \bar{S}_1}(x) = \mu_{S_1}(x) \wedge \mu_{\bar{S}_1}(x) = \left\{ \frac{0.5}{20} + \frac{0.35}{40} + \frac{0.15}{60} + \frac{0}{80} + \frac{0}{100} \right\}$

How to chose the membership functions

- ▶ Like in other aspects of the fuzzy sets theory, there are no clear “recipes” for choosing the membership functions of the fuzzy sets
- ▶ If we want to reduce the computations, we will prefer linear membership functions, i.e., triangles and trapeziums
- ▶ There are cases when we prefer non-linear membership functions (trigonometric, Gauss-type, etc):
 - ▶ There exist researchers that consider that linear membership functions do not provide the best results for some problems, while non-linear functions perform better
 - ▶ Sometimes the problem or the domain might need some types of membership functions
 - ▶ If we combine fuzzy sets theory with other methods, e.g., neural networks, it can be necessary to use membership functions that are suitable for these methods.

Introduction to Fuzzy Logic using MATLAB

Chapter 4 Membership Functions

- A ***membership function (MF)*** is a curve that defines how each point in the input space is mapped to a membership value (or degree of membership) between 0 and 1.
- The input space is sometimes referred to as the *universe of discourse*, a fancy name for a simple concept.

4.1 Introduction

- Fuzziness in a **fuzzy set is characterized** by its membership functions.
- It classifies the element in the set, whether it is **discrete or continuous**.
- The membership functions can also be formed by **graphical representations**.
- The graphical representations may include **different shapes**.
- There are certain restrictions regarding the shapes used.
- The rules formed to **represent the fuzziness** in an application are also **fuzzy**.
- The “**shape**” of the membership function is an important criterion that has to be considered. There are different methods to form membership functions.
- This chapter discusses on the features and the various methods of arriving membership functions.

4.2 Features of Membership Function

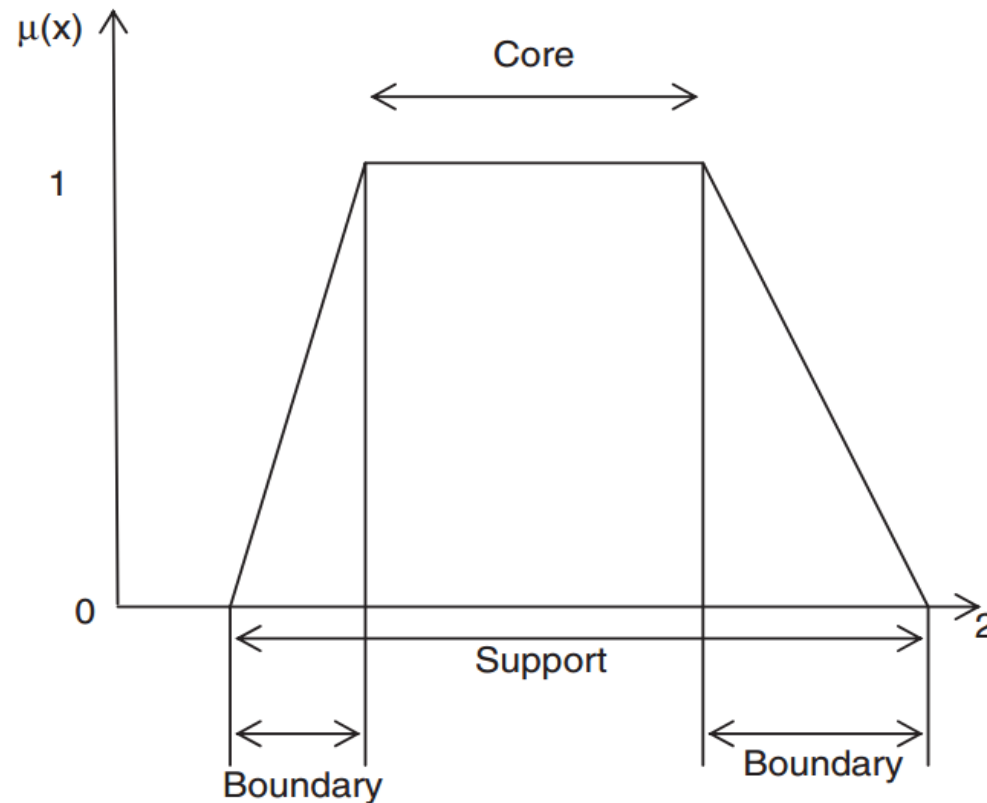


Fig. 4.1. Features of membership function

The feature of the membership function is defined by three properties. They are:

- (1) Core
- (2) Support
- (3) Boundary

The membership can take value between 0 and 1.

4.3 Classification of Fuzzy Sets

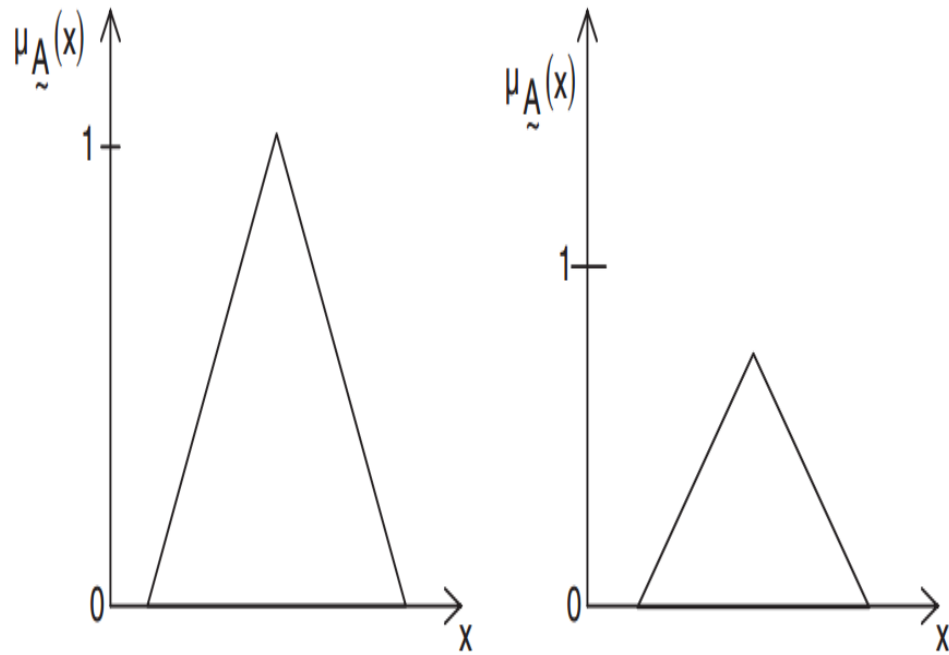


Fig. 4.2. (1) Normal fuzzy set and (2) subnormal fuzzy set

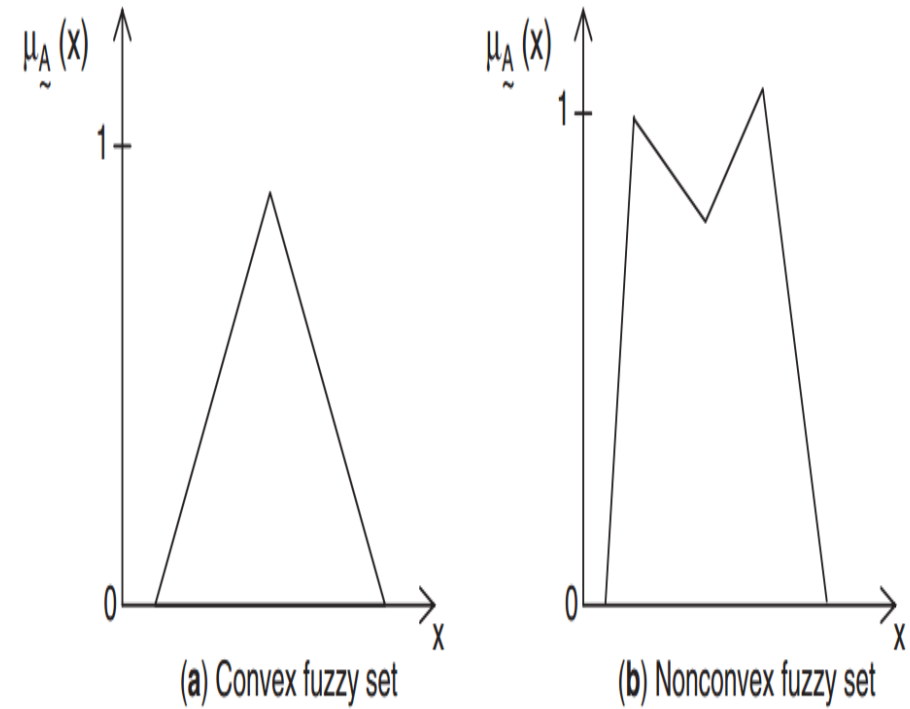


Fig. 4.3. (a) Convex set and (b) Nonconvex set

The fuzzy sets can be classified based on the membership functions. They are:

Normal fuzzy set. If the membership function has at least one element in the universe whose value is equal to 1, then that set is called as normal fuzzy set.

Subnormal fuzzy set. If the membership function has the membership values less than 1, then that set is called as subnormal fuzzy set.

These two sets are shown in Fig. 4.2.

Convex fuzzy set. If the membership function has membership values those are monotonically increasing, or, monotonically decreasing, or they are monotonically increasing and decreasing with the increasing values for elements in the universe, those fuzzy set $\tilde{A}$ is called convex fuzzy set.

Nonconvex fuzzy set. If the membership function has membership values which are not strictly monotonically increasing or monotonically decreasing or both monotonically increasing and decreasing with increasing values for elements in the universe, then this is called as nonconvex fuzzy set. Figure 4.3 shows convex and nonconvex fuzzy set.

The membership functions can have different shapes like triangle, trapezoidal, Gaussian, etc.

Design of membership function

Source:

membership function (pdf)

membership function2 (pdf)

4.4 Fuzzification

- **Fuzzification** is the process where the crisp quantities are converted to fuzzy (crisp to fuzzy).
- By identifying some of the **uncertainties present in the crisp values**, we form the fuzzy values.
- The conversion of fuzzy values is represented by the membership functions.
- In any practical applications, in industries, etc., measurement of voltage, current, temperature, etc., there might be a negligible error.
- This causes imprecision in the data. This imprecision can be represented by the membership functions. Hence fuzzification is performed.

FUZZIFICATION

Fuzzification is the process of changing a real scalar value into a fuzzy value.
This is achieved with the different types of **fuzzifiers (membership functions)**.

ZIFICATION

Fuzzy Linguistic Variables are used to represent qualities spanning a particular spectrum

Temp: {Freezing, Cool, Warm, Hot}

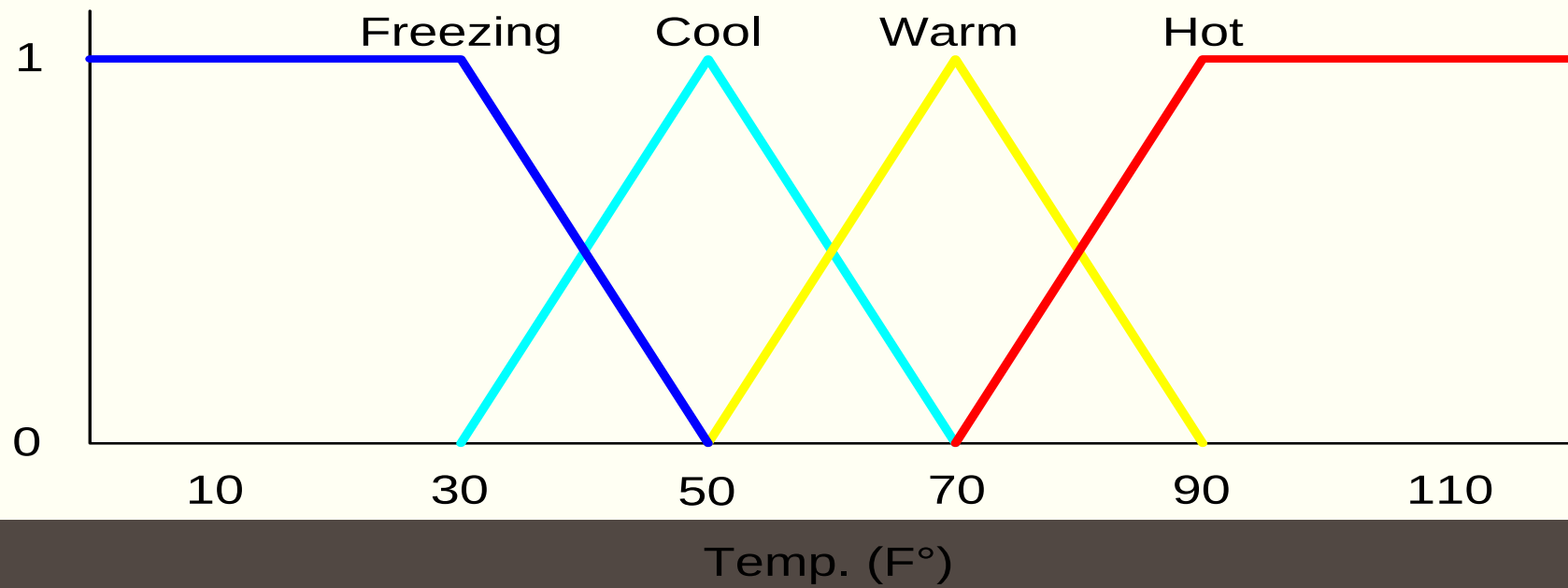
Question: What is the temperature?

Answer: It is warm.

Question: How warm is it?

ZIFICATION

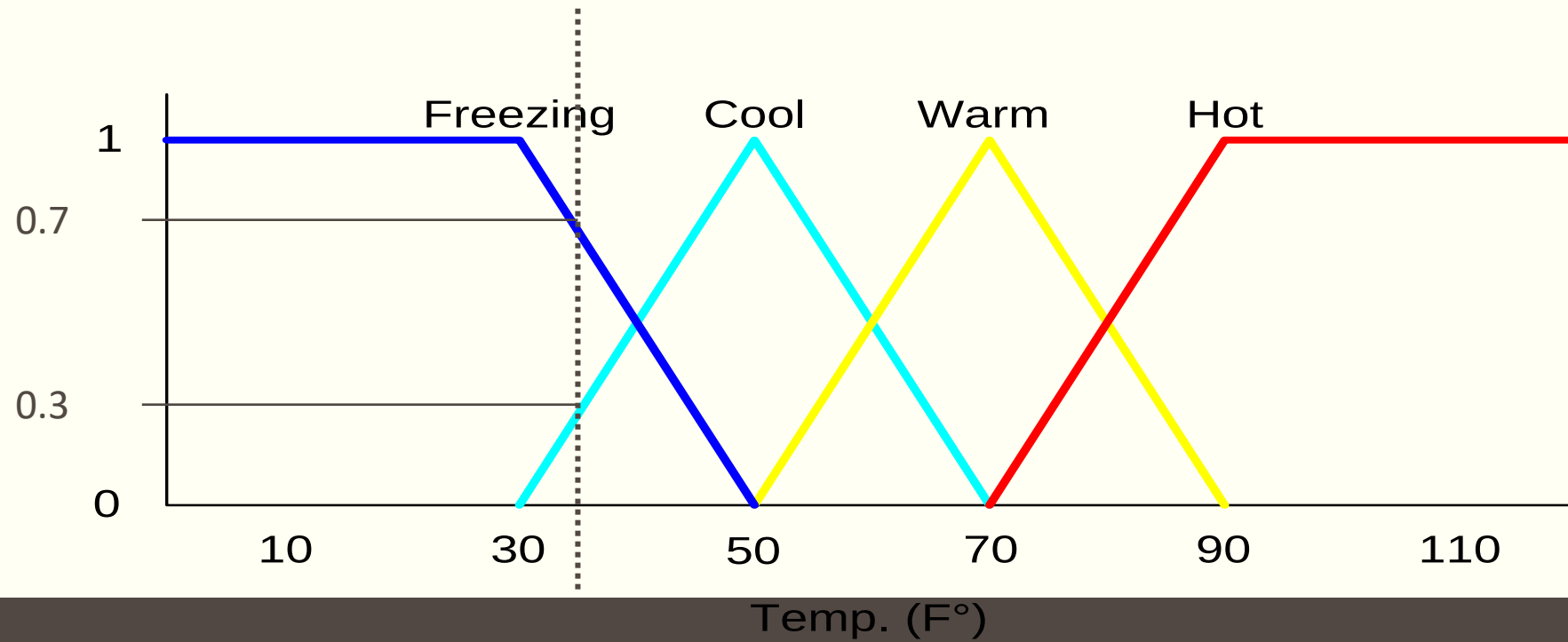
Temp: {Freezing, Cool, Warm, Hot}
Degree of Truth or "Membership"



ZIFICATION

How cool is 36 F° ?

It is 30% Cool and 70% Freezing



4.5 Membership Value Assignments

- The methods for assigning the membership values are listed as follows:
 - Intuition,
 - Inference,
 - Rank ordering,
 - Angular fuzzy sets,
 - Neural networks,
 - Genetic algorithms, and
 - Inductive seasoning

4.5.1 Intuition

Intuition is based on the human's own intelligence and understanding to develop the membership functions. The thorough knowledge of the problem has to be known, the knowledge regarding the linguistic variable should also be known. Figure 4.5 shows membership function for imprecision in crisp temperature reading.

For example, consider the speed of a dc-motor. The shape of the universe of speed given in rpm is shown in Fig. 4.6.

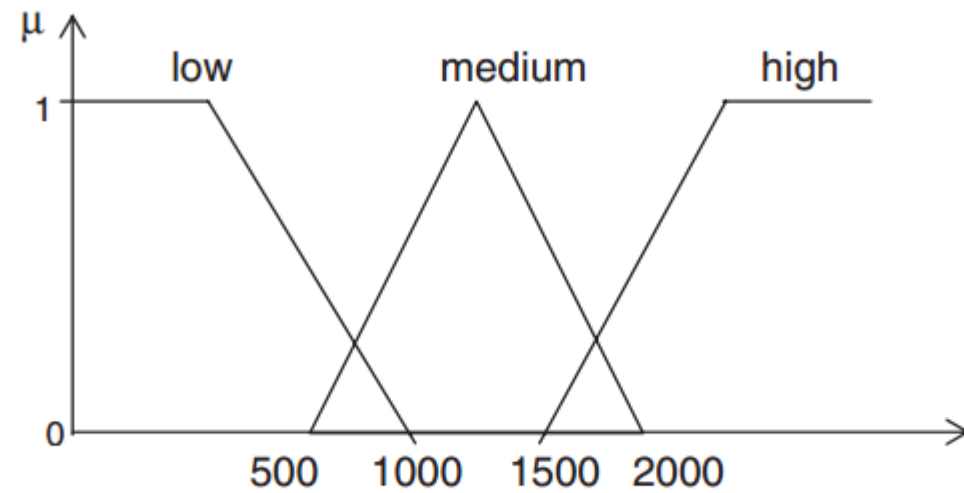


Fig. 4.6. Membership for fuzzy variable “speed” in rpm

Example 4.4. Using your own intuition and definitions of the universe of discourse, plot fuzzy membership functions for “weight of people.”

Solution. The universe of discourse is the weight of people.
in “kg” – kilogram.

Let the linguistic variables are:

Very light – $w \leq 30$

Light – $30 < w \leq 45$

Average – $45 < w \leq 60$

Heavy – $60 < w \leq 75$

Very heavy – $w > 75$

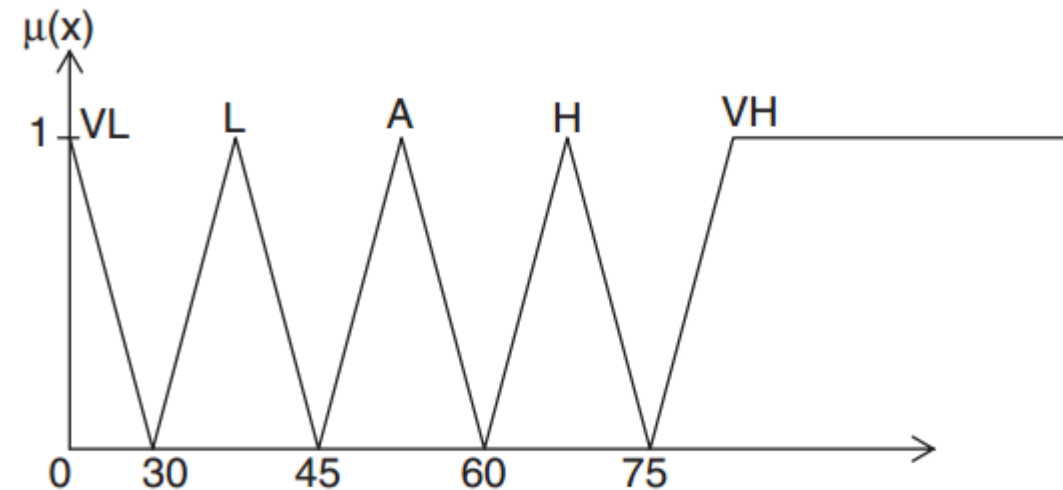


Fig. 4.12. Membership function of weight of people

Defuzzification

- Defuzzification means the fuzzy to crisp conversions.
- The fuzzy results generated cannot be used as such to the applications, hence it is necessary to convert the fuzzy quantities into crisp quantities for further processing.
- This can be achieved by using defuzzification process.



Source :

PDF Chapter 5 Defuzzification Methods